DEREK **KOCH**

Sussex, NJ | linkedin.com/in/derek-koch-7427161b6
973-896-2679 | derek@kochworx.com

Economics graduate (B.A., Cum Laude) with hands-on data analysis experience in Excel, SQL, Power BI, and R. Applies econometric methods (linear and logistic regression, difference-in-differences) to large public datasets, and has built Excel tools that automate calculations against code-based rate schedules. Brings a year of experience reconciling financial data, resolving discrepancies, and working within regulatory compliance requirements. Completing the Microsoft Office Specialist: Excel Expert certification. Comfortable learning new domains fast, open to travel and on-site client work.

SKILLS

- **Data & Analytics:** Advanced Excel (VLOOKUP/lookup and logical functions, automated estimating models, data cleaning), SQL (table creation, SELECT queries) in SQL Server/SSMS and Azure SQL, Power BI, R Studio, Python
- **Statistical Methods:** Linear regression, logistic regression, difference-in-differences, data visualization
- **Compliance & Process:** Code-based fee and rate schedules, regulatory research, documentation and submission workflows, HIPAA compliance
- **Tools & Methods:** Microsoft Office Suite (Word, Excel, PowerPoint, Outlook), Azure DevOps, Agile/Scrum

RESEARCH & ANALYTICS PROJECTS

Independent Studies

2025 – 2026

AML Transaction Monitoring Analysis (Excel) | Oct 2026

- Built AML transaction-monitoring checks in Excel on sample transaction, customer, and watchlist data to prepare for an AML Data Analyst Excel skills assessment (passed)
- Flagged potential structuring (cash deposits of $9,000–$9,999), CTR-threshold activity, round-dollar amounts, and duplicate transaction IDs using COUNTIFS, SUMIFS, MOD, and nested IF helper columns
- Screened customer names against a watchlist with TRIM-cleaned COUNTIF matching, and pulled customer risk ratings across sheets with VLOOKUP and INDEX/MATCH to total high-risk activity
- Summarized activity by customer, transaction type, and month with PivotTables, date grouping, and conditional formatting

2008 Financial Crisis & Household Housing Recovery (R Studio) | 2025, William Paterson University

- Designed and ran an independent study using Federal Reserve Survey of Consumer Finances (SCF) data in R Studio
- Cleaned and modeled the data using difference-in-differences, linear, and logistic regression to evaluate the impact of post-crisis borrowing-cost relief across income brackets
- Found recovery after 2009 concentrated among households earning $350K+, with lower-income brackets taking significantly longer despite federal policy efforts
- Presented findings and data visualizations at the statewide New Jersey Big Data Alliance (NJBDA) Research Symposium

Honors Thesis: Business Plan for a Wealth Management Firm

Aug 2024 – May 2025

- Led industry analysis for a 7-person team, estimating market size and projected growth to support the plan's financial projections
- Researched federal securities laws and regulatory frameworks affecting operations; presented and defended recommendations to faculty and industry professionals

WORK EXPERIENCE

Receptionist / Assistant Biller

Sep 2025 – Oct 2026

Alberto and Cho Oral and Maxillofacial Surgery Group, LLC | Vernon & Sparta, NJ

- Built an Excel estimating tool using VLOOKUP and IF logic that pulls rates from carrier fee schedules by procedure code and recalculates patient estimates automatically, cutting revision time by 10–15 minutes per estimate
- Reconciled insurance payments against billed amounts across 5 carriers (Cigna, Aetna, MetLife, BCBS, Delta Dental) for a high-volume practice seeing 10–30 patients daily
- Analyzed EOBs to identify denial reasons and discrepancies, then corrected and resubmitted claims and predeterminations with supporting documentation through NEA FastAttach
- Calculated expected out-of-network reimbursement using carrier portals to resolve high-balance billing disputes
- Worked across patients, carriers, and referring practices to coordinate scheduling, coverage, and documentation
- Trained a new receptionist on insurance verification, billing workflows, and the estimating tool

- Maintained HIPAA compliance in handling protected health information

Park Associate II (promoted from Cashier) 2023 – 2025
LEGOLAND New York | Goshen, NY

- Promoted from Cashier to Park Associate Tier 2; cross-trained across attractions, retail, and park operations
- Processed high-volume cash and card transactions and followed safety and operating procedures in a fast-paced setting

Additional Experience: Cashier, ShopRite (Feb – Mar 2025, seasonal) | Cashier, Lowe's (2022 – 2023)

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

- **Microsoft Office Specialist: Excel Expert** (in progress) | **Excel 2019 Associate** (2025)
- **Power BI for Data Analysts** coursework (in progress) | **SQL Fundamentals**, Pluralsight | **Scrum Fundamentals**
- **The Institute of Internal Auditors (IIA)**, Member (2026) | **Associate in General Insurance (AINS)** (in progress)

EDUCATION

Bachelor of Arts, Economics (Business Honors) May 2025
William Paterson University, Honors College | Wayne, NJ

- Cum Laude | GPA: 3.46 | Dean's List (2 semesters) | Honors Scholarship recipient
- Relevant Coursework: Applied Data Analytics (case-based problem solving in R Studio); Research-Based Industry Analysis (independent study design and econometric analysis)

EXTRACURRICULAR & VOLUNTEERING

- **Data Analytics & Student Investment Club:** Member (Sept 2024 – May 2025); sessions on Excel and Tableau
- **Leukemia & Lymphoma Society, Team in Training:** Raised ~$1,500 with a family fundraising team; completed the Marine Corps Marathon and Liberty City Half Marathon